Reducing Space Junk:
A Necessity for Our Future

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Debris Zero

Abstract

The rapid accumulation of space debris poses an escalating threat to global technological infrastructure, further orbital endeavors, and economic viability. Orbital debris or "space junk" endangers current and future space missions, particularly as satellite launches increase worldwide. A critical concern is the potential onset of Kessler Syndrome, a self-perpetuating cascade of collisions that could render key orbital paths unusable. Economically, collision avoidance maneuvers, satellite losses, and insurance risks significantly inflate operational costs, stifling innovation and access to space. Addressing this complex issue requires a multifaceted strategy, including active debris removal technologies, preventive satellite design, and higher funding in the space. Coordinated global action is essential to ensure long-term orbital sustainability and protect space as a shared resource for communication, navigation, defense, and scientific advancement.

1 The Problem: A Crisis in Orbit

The proliferation of space debris, commonly known as "space junk," represents an increasingly severe global crisis with profound implications for our society, technology, and economy. Space junk includes defunct satellites, spent rocket stages, collision fragments, and other discarded objects orbiting Earth at velocities up to tens of thousands of kilometers per hour. As more nations and private companies launch satellites and spacecraft, the volume of orbital debris continues to grow exponentially, with over 36,000 objects larger than 10 cm tracked by agencies today, and millions of smaller untracked fragments. These objects provide no utility but pose continuous and escalating threats to the satellites that underpin global communications, navigation, weather forecasting, and national security. The most catastrophic potential outcome is Kessler Syndrome, first identified by NASA scientists Donald J. Kessler and Burton G. Cour-Palais in 1978 [1]. This theoretical but increasingly plausible scenario describes a self-sustaining cascade: each collision between debris creates thousands of new fragments, which raise the probability of further collisions, which create more

fragments. Over time, this runaway cycle could render entire orbital bands effectively unusable for decades or centuries. Beyond the existential risk of Kessler Syndrome, the day-to-day economic toll is already substantial. Satellite losses and replacement costs run into the billions, collision-avoidance maneuvers strain operational budgets, and insurance premiums for new launches continue to rise. Even minor disruptions to satellite services can trigger cascading failures in sectors dependent on precise timing and positioning—including aviation, financial trading, and emergency response. Uncontrolled re-entries of large debris further endanger populations on Earth, as demonstrated by the unpredictable trajectories of China's Long March 5B rocket stages in recent years [5]. Left unaddressed, this problem will only compound.

2 The Reaction: Efforts Underway—and Why They Fall Short

The global space community has not been idle. Over the past decade, a growing recognition of the debris problem has prompted a wave of policy initiatives, technical proposals, and international agreements. The United Nations Committee on the Peaceful Uses of Outer Space (COPUOS) established voluntary debris mitigation guidelines in 2007, and major space agencies—including NASA, ESA, and JAXA—have adopted internal policies requiring new satellites to de-orbit within 25 years of mission completion. Several nations have begun mandatory licensing requirements that include debris mitigation plans as a condition for launch approval. On the technology front, proof-of-concept missions have demonstrated early promise. The RemoveDEBRIS mission (2018) successfully tested a net-capture system in orbit. ESA's ClearSpace-1 mission, currently in development, is designed to capture and de-orbit a derelict rocket adapter—marking what would be the first commercial active debris removal operation. Japan's Astroscale has launched its ELSA-d mission to test magnetic docking for debris capture. These are meaningful steps forward. However, the current response remains critically insufficient relative to the scale and pace of the problem. The 25-year de-orbit rule is voluntary and widely ignored. Compliance rates among commercial operators remain low, and no binding international mechanism exists to enforce it. The commercial launch boom, driven by mega-constellations from operators like SpaceX (Starlink), Amazon (Kuiper), and OneWeb, is adding thousands of new objects to orbit far faster than any cleanup effort can address. Legacy debris from decades of prior launches, including thousands of defunct satellites and rocket bodies, sits completely unaddressed by current frameworks. In short, the reaction to date has produced awareness and early tools, but not the coordinated, funded, and enforceable global action that the problem demands. A new, more decisive phase of response is urgently needed. [3]

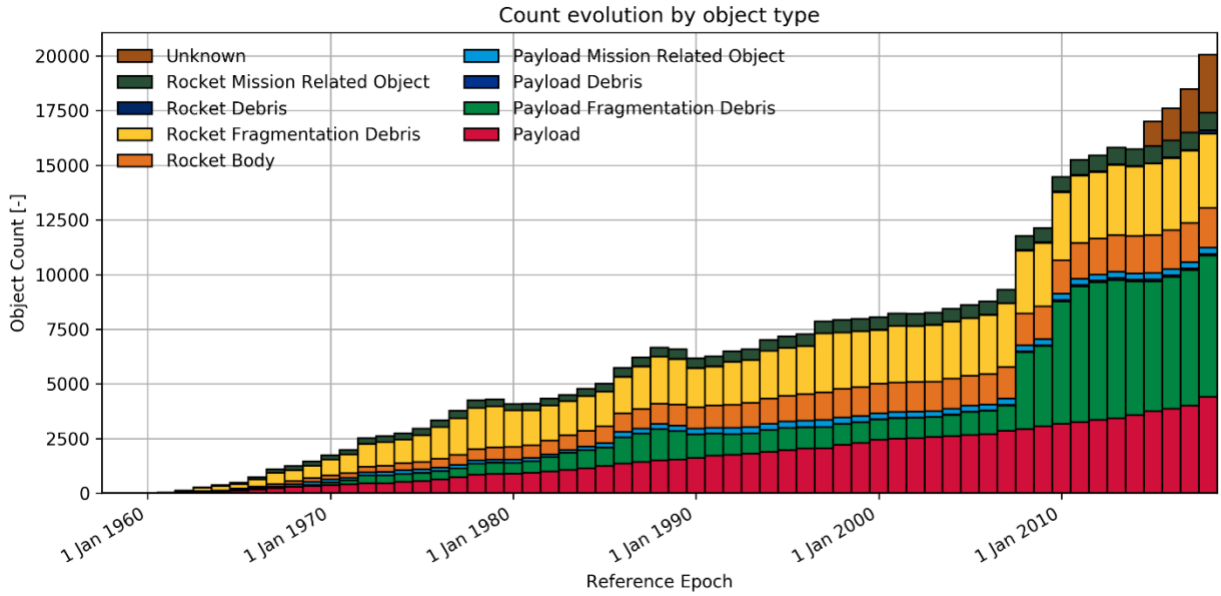

Figure 1: A graph showing the object count of space debris in or above low earth orbit. [2]

3 The Solution: A Coordinated, Three-Pillar Response

Addressing the space debris crisis requires more than incremental improvements to existing practices. What is needed is a sustained, internationally coordinated strategy built around three core pillars: active debris removal, preventive design standards, and binding global governance. Each pillar is necessary; none is sufficient on its own.

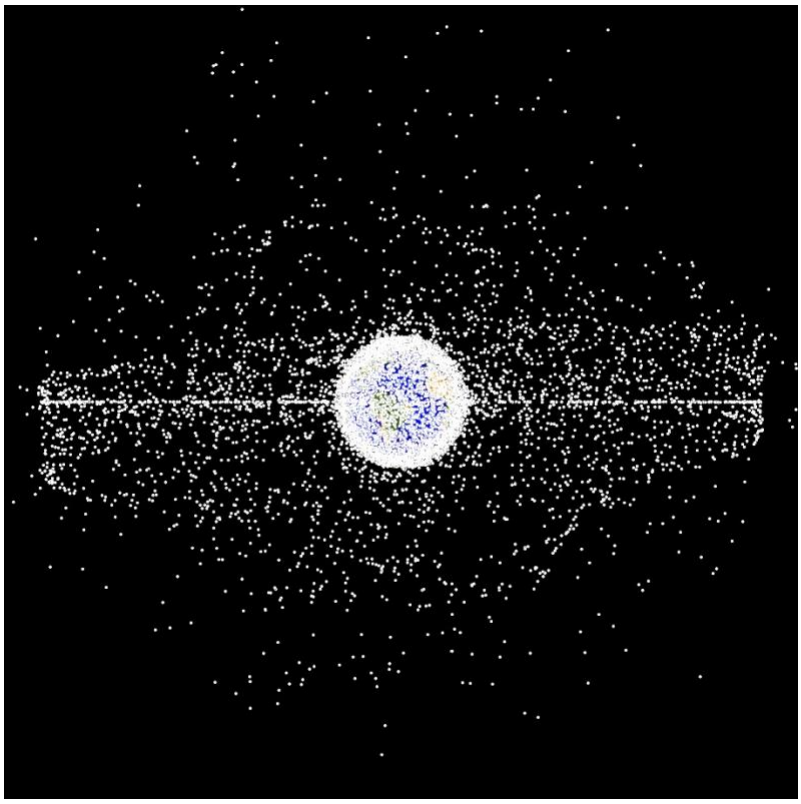

Figure 2: Render of all known large space debris according to the NASA Orbital Debris Program Office [6]

Pillar 1: Active Debris Removal

Active debris removal (ADR) missions involve specialized spacecraft designed to capture, stabilize, and either de-orbit or relocate dangerous debris into safer trajectories. Projects such as ClearSpace-1, developed in collaboration with the European Space Agency (ESA), aim to demonstrate practical debris removal by targeting large derelict satellites or rocket stages for controlled re-entry. Technologies like robotic arms, harpoons, nets, and magnetic docking systems have been proposed and tested to capture and safely remove debris from critical orbits [8]. The most urgent priority should be targeting the roughly 50 largest objects in low Earth orbit: defunct rocket stages and satellites that pose the greatest collision risk and would generate the most new fragments if struck. Governments and space agencies should commit to funding at least five dedicated ADR missions per year beginning in 2026, with a goal of removing the 100 highest-risk objects by 2035.

Figure 3: Clearspace-1's mission plan detailing the strategy used to safely deorbit debris [7]

Pillar 2: Preventive Design Standards

New satellites must be built to leave no lasting footprint. Enhanced design mandates should require built-in propulsion systems capable of de-orbiting within 5 years of mission completion—a sharp tightening of the current voluntary 25-year guideline. Spacecraft should be required to carry sufficient reserve fuel to execute a controlled re-entry or graveyard orbit transfer at end of life. These requirements should apply to all operators seeking launch licenses from any signatory nation, eliminating the race-to-the-bottom dynamic that currently allows non-compliant operators to launch freely [9].

Pillar 3: Binding International Governance

The current voluntary framework has demonstrably failed to keep pace with the commercial launch boom. The United Nations Office for Outer Space Affairs (UNOOSA) should be empowered to develop a binding treaty that establishes enforceable debris mitigation requirements, a global debris tracking and transparency regime, and a liability and remediation fund financed by launch operators proportional to their orbital footprint. Advanced tracking systems and AI-driven predictive modeling should be developed as shared international infrastructure, enhancing debris detection and collision avoidance for all operators. Education and public awareness campaigns can further build the political will needed to sustain this governance push over the long term [10].

4 Conclusion

The space debris crisis is no longer a distant theoretical concern, it is an active and worsening threat to the satellite infrastructure that modern civilization depends on. The global community has recognized the problem and taken early steps, but those steps have been too slow, too voluntary, and too fragmented to match the pace of the challenge. The three-pillar approach outlined in this paper: active debris removal, preventive design standards, and binding international governance, offers a coherent and actionable path forward. None of these pillars requires technologies that do not yet exist. What they require is political will, coordinated investment, and a shared commitment among space-faring nations and private operators to treat orbital sustainability as the common responsibility it is. The window to act decisively is narrowing. Kessler Syndrome, once triggered, cannot be reversed. Implementing these solutions now is not merely a matter of protecting space assets—it is a necessity for securing humanity's continued access to space, and the countless benefits that access provides, for generations to come.

References

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